OPH: how records become space, clocks and testable physics

IPI discussion note | 2 November 2026 | Intuition, finite results and executable-model audit

OPH's central idea is to recover physical descriptions from what local observers can record, compare and change. Its appeal is a concrete economy: the same interface records can describe a local change, support a clock and help reconstruct geometry. The finite constructions and linked field theories make parts of this idea precise. Whether they describe our measured universe remains an empirical question. This note explains the construction before stating its dispersion and direct-test commitments.

Author's position. The twelve-port carrier is absolutely essential to OPH: it is what constrains space to three dimensions and fixes the Standard Model symmetries. If the proposed carrier class is excluded, I do not think OPH would survive. Here “carrier class” refers to the core construction; a test of one optional dispersion law has narrower scope.

1. Space starts with relations, rather than assigned coordinates

What is an observer? A subsystem with access to some records and operations, not necessarily a conscious subject. Two observers can see different parts of a process. Agreement means that, when their records refer to the same shared event or quantity, their interpretations must match under the allowed comparison maps. It does not require identical private histories. OPH assumes a carrier architecture with twelve boundary ports, thirty pairwise seams and twenty faces, combinatorially an icosahedron; it does not derive that architecture from nothing. [1]

What happens locally? In the elementary executable model, ports carry integer loads. A seam repair balances its endpoints as evenly as possible; a fair coin chooses between the two orientations when their total is odd. Thus $(1, 0)$ becomes $(1, 0)$ or $(0, 1)$ with equal probability. The event records its signed transferred load j. These loads are model variables, not already identified physical particles. Different admissible quantum record laws preserve different coherences while sharing this classical rule. [9]

A record therefore says more than “an update occurred”: it says where a change was exchanged. With the initial local load known, an observer reconstructs its later value by adding its incident currents:

$$n_i(N) = n_i(0) + \sum_{s \le N} j_s(\mathbf{1}_{i=b_s} - \mathbf{1}_{i=a_s}).$$

The two ends record opposite changes. Overlapping observers can reconstruct the same shared values without each reading the complete state. The code checks this identity on all 3,960 binary total-two branches and executes continued local readouts. Initial-state disclosure and complete incident records are required. [2]

How does “near” become a distance? Directly coupled ports are neighbors in the supplied incidence graph. To go beyond that graph, perturb the load and observe its relaxation. The slow modes are patterns that survive the fastest local equalization. Under uniform seam choice the mean one-step response is $M = I - L/60$. The icosahedral Laplacian L has a slow nonconstant eigenspace of dimension three, at eigenvalue $5 - \sqrt{5}$. Its projector P yields the selected unit Gram matrix $G = 4P$: the entries are dot products of twelve vectors in three dimensions. Their distances are

$$d_{ij}^2 = G_{ii} + G_{jj} - 2G_{ij}.$$

Thus three-dimensional relative geometry is read from a response operator. Coordinates can be factored from G, but rotating the entire coordinate system changes no distance. This is an exact finite result on the selected response, not a deduction of a physical continuum merely from the number three. The graph is architectural input; its record-based response and Gram geometry are recovered. [2]

What is relative position? At this level it is a pattern of distinguishable relationships to other ports or regions: which interfaces connect them, how perturbations propagate, and the resulting Gram distances. A chosen observer port sees graph-distance shells of sizes $1, 5, 5, 1$; this finite radial readout is recovered from currents. It is not an extra independent spatial dimension. In the extended construction, positions label localized field readouts, while their response relations determine the effective metric. An origin and axes are descriptions of those relations, not additional observable objects. [2,8]

2. Motion, local clocks and the recovered 3+1 description

Moving means transporting an identifiable pattern. If an excitation initially localized near A later changes the readouts near B, with its charge or quantum information tracked, the description assigns it motion. What matters is its changing relations to the observer, not a change of coordinate labels. The elementary current is an exact transfer; coherent propagation transports amplitudes and correlations. Executable examples transport occupations and an actively protected code whose location follows recorded currents. That code needs recovery and eventually stops; it is not yet a stable autonomous particle. A moving observer would carry its record history while its neighbors change. Switching read ports is implemented, but alone does not establish that physical worldline. [2,8,11]

Time starts with ordered change. The executable process has an update order; dependencies among records identify which events can influence later ones. A local observer need not see every update. One legitimate clock counts every attempted repair on an incident seam, including attempts with zero current:

$$N_i(n) = \sum_{s=1}^{n} \mathbf{1}_{\{i \in e_s\}}, \qquad \mathbb{E}[N_i(n)] = n/6$$

for uniform choice among thirty seams of an isolated carrier. There are five seams at each port. Six global attempts per local tick is an average, not a rule that every sixth attempt reaches that observer. A clock counting only actual movements is a different, state-dependent clock. “Observer-subjective” means tied to the observer's accessible events, not arbitrary or psychological. [2,11]

Different clocks can still be compared. In a retained movement-count example, two observers read $(1, 1)$ and then $(2, 2)$ at shared events, but later $(5, 3)$: one saw two additional local movements. The first two coincidences do not justify declaring their clocks identical forever. Finite clock theorems fix a positive affine comparison from two shared events and test it on a third. On refining screens, exposure-normalized incident counts converge to the same dimensionless time parameter under the specified scheduler; predictable changes of read port have a proved degree-compensated version. Clock and field fluctuations are treated jointly. These results do not identify every local count with relativistic proper time. [2,11]

The field link really has been constructed. Many joined carriers yield a native screen operator A of spectral dimension two. It measures how strongly a spatially varying field pattern is penalized. A declared radial extension gives $B = -\partial_r^2 + (A \oplus 0)$. This same product operator fixes field covariance, thermal correlations and wave response, with source-compiled approximations to coherent frequencies $\sqrt{B}$. With the specified hyperbolic action, its spatial energy reconstructs a measurable metric of signature $(-, +, +, +)$. There are also source-executed three-dimensional scalar regulators and a specified Weyl/contact branch with a free eight-valley relativistic continuum. These are established conditional constructions, not missing wholesale. [8]

This gives relative distance an operational meaning beyond drawing points: for the declared product wave, the infimum of times at which a complete local probe in one region can affect another equals their intrinsic optical distance. The wave operator and metric agree. A particular detector can miss the signal; the statement ranges over the complete local response. Finite spectral cutoffs need not have an exact front. The radial design and coherent wave protocol remain inputs. [8]

What do the finite spacetime theorems add? They prove that compatible event charts transform relative displacements consistently. On a supplied unit-timelike affine history, the clock increment satisfies $(\Delta\tau)^2 = (\Delta t)^2 - |\Delta \mathbf{x}|^2$ in units $c = 1$, independently of the overlapping chart. They do not manufacture that chart or calibration from bare ticks. A faithful reference state also defines modular generator $K = -\log \rho$; a specified Gibbs realization links its flow to Hamiltonian flow. A dimensionless evolution generator G becomes energy only after calibration: $H = (\hbar/T)G$ when $t = T\tau$. [11]

The achievement is a linked reconstruction of records, geometry, clocks and fields. The remaining task is to select and physically validate the radial, action, reference and calibration inputs together. A recovered 3+1 field description does not by itself force one particular microscopic dispersion law.

3. The frozen dispersion relation: prediction, intuition and exclusion

Additional physical hypothesis. A real field has positive difference-energy terms on icosahedrally invariant displacements and an ordinary quadratic time-kinetic term. A plane wave turns each difference into $1-\cos(\mathbf{k}\cdot\mathbf{d})$; therefore this wave dynamics has $\omega^2 = c_0^2\Lambda$. Passive repair diffusion is a different continuation of a spatial operator.

For complete orbits O_s of unit directions, positive dimensionless radii r_s, positive relative weights w_s, common orientation and $\mu_m = \sum_s w_s |O_s| r_s^m$, the complete stipulated spatial symbol is

$$\Lambda(\mathbf{k}) = \frac{6}{a^2\mu_2}\sum_s w_s \sum_{\mathbf{u}\in O_s}[1-\cos(ar_s\mathbf{k}\cdot\mathbf{u})], \qquad \omega^2 = c_0^2\Lambda.$$

Here a is in metres, k in m^{-1}, ω in s^{-1} and c_0 in $\mathrm{m\,s}^{-1}$. For an identified quantum mode, $p=\hbar k$ and $E=\hbar\omega$. Extra spatial terms, a momentum-dependent time-kinetic term or a different field identification change this hypothesis. [3,7]

With twelve unit vertex directions $\mathbf{v}_i$, define $I_6(\mathbf{n}) = (25/132)\sum_i P_6(\mathbf{v}_i\cdot\mathbf{n})$, where $P_6(x) = (231x^6 - 315x^4 + 105x^2 - 5)/16$. Its range is $[-5/9, 1]$. Then

$$\Lambda = k^2 + C_4k^4 + [B_0 + B_6 I_6(\mathbf{n})]k^6 + R_8, \quad C_4 = -\frac{a^2\mu_4}{20\mu_2}, \quad B_0 = \frac{a^4\mu_6}{840\mu_2}.$$

$$C_4 < 0, \qquad \frac{B_0}{C_4^2} \geq \frac{10}{21}, \qquad -\frac{16}{135} \leq \frac{B_6}{B_0} \leq \frac{16}{75}.$$

C_4 has units m^2 and B_0, B_6 have units m^4; the ratios are dimensionless. Angular ranks 1-5 vanish; rank six has one shape, with one common orientation in $SO(3)/A_5$. This is angular structure, not particle spin.

Unit-radius branch	C_4/a^2	B_0/a^4	B_6/a^4	B_6/B_0
Twelve vertices, FZ-11	−1/20	1/840	2/7875	16/75
Thirty seam directions, FZ-12	−1/20	1/840	−1/12600	−1/15
Twenty face directions	−1/20	1/840	−2/14175	−16/135

Why this shape? Symmetry makes low-order directional moments isotropic; its first allowed anisotropy is rank six. The cosine expansion fixes the negative quartic term. Positive weights give $\mu_2\mu_6 \geq \mu_4^2$ and bound the anisotropic weighted average. Equal radii saturate the floor. A positive vertex/face mixture with per-direction weights $25:27$ cancels the rank-six term, so nonzero anisotropy is not compulsory. [3]

Range and numerical decision. Require $ar_{\max}k \ll 1$ or explicit remainder control: $|R_8| \leq a^6\mu_8 k^8/(6720\mu_2)$; FZ-12 has $|a^2R_8| \leq 7(ak)^8/388800$ for $ak \leq 1$. Thus the low-energy condition is approximately $E \ll \hbar c_0/(ar_{\max})$, with no justified numerical eV range yet. Fit the full symbol or bound the longest hop uniformly. After nuisance and orientation profiling, a simultaneous confidence region disjoint from the class at global error probability $\leq 2.87\times10^{-7}$ is a sufficient five-sigma exclusion. For example, an invented, fully calibrated Gaussian result $B_0/C_4^2 = 0.400 \pm 0.010$ has five-sigma upper end $0.450 < 10/21$. [4]

A resolved violation rejects every member of this positive scalar cosine class. A null does not: effects can shrink arbitrarily, and anisotropy can cancel. A null requires an independently established lower physical hop scale and adequate power. Failure reaches all OPH only if this propagation law is proved necessary and exclusive; twelve ports and the existing field reconstruction do not establish that. [4]

4. Direct tests of the construction, and a laboratory path

A carrier signature without an unknown length. Under uniform seam choice and common state-independent idling, every native record law sharing the classical repair has $M = I - \eta L/60$, $0 < \eta \le 1$. Nonzero deficits $1 - \lambda(M)$ have multiplicities $3, 5, 3$ and ratios $1 : 3(5 + \sqrt{5})/10 : (3 + \sqrt{5})/2$. Twelve calibrated diagonal preparations and population readouts reconstruct the full map. This tests the specified native carrier and schedule; it is not yet a universal prediction for any system labelled "twelve-port." [9]

An interference test of the whole classified record-law family. Prepare $(|X\rangle + e^{i\theta}|Z\rangle)/\sqrt{2}$, with occupied ports $X = \{2, 3\}$ and $Z = \{1, 2\}$; repair seam $(0, 1)$ and read the phase-zero projector. Every seam-local, phase-covariant law in the family obeys

$$p_+(\theta) = \tfrac{3}{8} + \tfrac{1}{2}\mathrm{Re}(g e^{-i\theta}), \qquad p_+(0) + p_+(\pi) = \tfrac{3}{4}, \qquad |g| \le \tfrac{1}{\sqrt{2}}.$$

The unknown coherence cancels in the sum. A calibrated violation excludes that entire family, not merely C versus E. A programmable ion or photonic processor can validate the encoded experiment. Programming the rule and seeing it work tests the implementation; testing nature requires an independent identification of the physical preparation, repair and readout. All trials, aborts and apparatus errors must be included. [9,10]

A route closer to the twelve-port axiom itself. A1 makes the complete reversible port response a commutator-closed image of $\mathbb{R}^{12}$. For preparation/readout pairs (ρ_a, E_a) and infinitesimal port-generated operations X_b, response columns factor through those twelve generators:

$$J_{ab} = \mathrm{Tr}(E_a[X_b, \rho_a]), \quad \mathrm{rank}\, J \le 12; \quad \sigma_{13}(\hat{J}) > \varepsilon \implies \text{reject}$$

if $\|\hat{J} - J\|_2 \le \varepsilon$ with preregistered simultaneous coverage. This elementary consequence of A1 has exact finite controls in the audit, not a completed physical experiment. The bound includes derivative approximation, drift, irreversibility and tomography errors. Finite-angle response matrices can have higher rank; only the infinitesimal matrix has this bound. [1,9]

Proposed laboratory programme. On a candidate single carrier, address twelve putative port controls, use diverse preparations and binary readouts, and vary signed pulse amplitudes near zero. Add commutator loops of those same controls to seek a thirteenth independent direction. Gate-set tomography can treat preparation and readout errors jointly. Freeze the tested sector, probes, derivative bounds and decision rule before held-out runs. A certified thirteenth direction excludes the closed twelve-dimensional response independently of dispersion and scale. [10]

The decisive prerequisite is identifying this as **one primitive carrier's complete port response**. Thirteen knobs or universal encoded gates might use multiple carriers or ancillary systems. OPH must fix that mapping independently of the result; the current field reconstruction does not settle it. The optical alternative is a rotating, multi-frequency vacuum-cavity test of the conditional dispersion. Its angular fractional frequency term is $B_6 k^4 I_6/2$; a Planck-length FZ-12 example at 1064 nm is only about 5×10^{-117}. Direct carrier tests avoid that suppression but require more specific operational access. [3,6,7]

Audit status. The July/August frozen rays and the later class theorem were kept distinct. Four original target/prediction hashes, coefficients, positive mixtures, signed countercontrols, cancellation and remainders were checked. Across the dispersion, direct-field and clock/motion audits, 369 targeted tests passed, with four independent verifiers. Separate exact controls check 3,960 native branches and the proposed response-rank witness. All commands are logged. No new Lean kernel rebuild, external timestamp verification or empirical confirmation is claimed. [5,9]

Sources and reproducibility

[1] OPH axioms. [2] Observer-relative reconstruction. [3] Dispersion certificate and adjacent code. [4] Frozen register.

[5] Audit and verification log; hashes identify working-tree inputs, not necessarily baseline-commit bytes. [6] Cavity methodology. [7] NIST constants. [8] Linked fields and clocks, and its linked product, scalar and Weyl constructions.

[9] Direct-test audit, model/record_laws/. [10] Gate Set Tomography. [11] Finite clock comparisons, adjacent proper-time theorem, model/moving_defect/ and model/defect_field_propagation/.